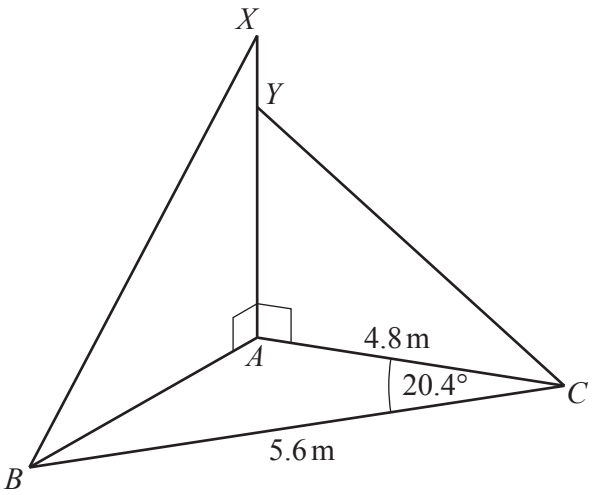
0580_s23_qp_43_Q5

5 (a)



NOT TO SCALE

ABC is a scalene triangle on horizontal ground.
AYX is a straight vertical post, held in place by two straight wires *XB* and *YC*.
AC = 4.8 m, *BC* = 5.6 m and angle *ACB* = 20.4°.

(i) Calculate *AB*.

AB = ... m [3]

(ii) Angle *XBA* = 64°.

Calculate *AX*.

AX = ... m [2]

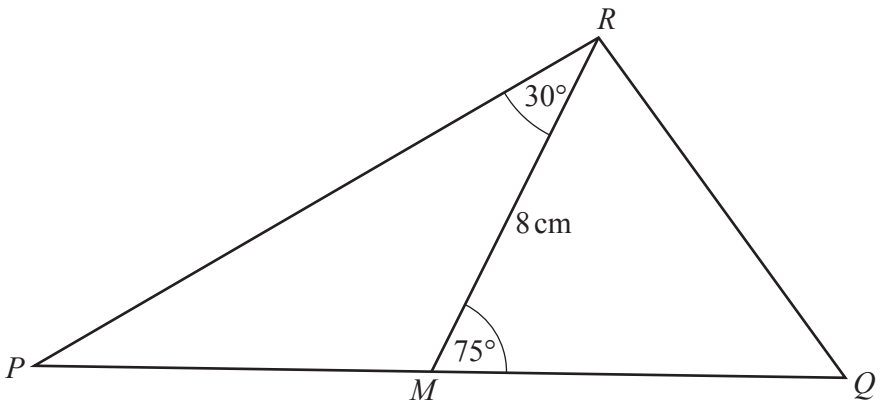
(iii) *AY* = 2.9 m.

Calculate the area of triangle *YAC*.

..... m^2 [2]

0580_s23_qp_43_Q5

(b)



NOT TO SCALE

In triangle PQR, M is the midpoint of PQ.
$RM = 8$ cm, angle $PRM = 30°$ and angle $RMQ = 75°$.

Calculate PQ.

$PQ =$ ..... cm [5]